

## Assignment No.9

```
test> db.student.insertMany([
... { _id: 101, Name: "Pratik", Class: "SE", Div: "B", City: "Pune", Marks: { TOC: 44, DBMS: 33,
CN: 57, SEPM: 67 } },
... { _id: 102, Name: "Sumeet", Class: "SE", Div: "A", City: "Mumbai", Marks: { TOC: 55,
DBMS: 61, CN: 62, SEPM: 72 } },
... { _id: 103, Name: "Suyash", Class: "SE", Div: "B", City: "Pune", Marks: { TOC: 35, DBMS:
28, CN: 47, SEPM: 54 } },
... { _id: 104, Name: "Ankita", Class: "SE", Div: "A", City: "Pune", Marks: { TOC: 48, DBMS:
52, CN: 58, SEPM: 65 } },
... { _id: 105, Name: "Kadambari", Class: "TE", Div: "C", City: "Mumbai", Marks: { TOC: 60,
DBMS: 45, CN: 55, SEPM: 70 } },
... { _id: 106, Name: "Pragati", Class: "TE", Div: "B", City: "Pune", Marks: { TOC: 50, DBMS:
40, CN: 60, SEPM: 68 } },
... { _id: 107, Name: "Rohit", Class: "TE", Div: "B", City: "Mumbai", Marks: { TOC: 72, DBMS:
55, CN: 65, SEPM: 74 } },
... { _id: 108, Name: "Sneha", Class: "TE", Div: "A", City: "Pune", Marks: { TOC: 66, DBMS:
59, CN: 70, SEPM: 77 } }
... ] );
{
  acknowledged: true,
  insertedIds: {
    '0': 101,
    '1': 102,
    '2': 103,
    '3': 104,
    '4': 105,
    '5': 106,
    '6': 107,
    '7': 108
  }
}
```

#### **// Q1. Average TOC marks of TE students**

```
test> db.student.aggregate([{$match:{Class:"TE"}}, {$group:{_id:null, avgTOC:{
$avg:"$Marks.TOC"}}}]);
```

```
[ { _id: null, avgTOC: 62 } ]
```

#### **// Q2. Min DBMS marks of TE Div B students**

```
test> db.student.aggregate([{$match:{Class:"TE", Div:"B"}}, {$group:{_id:"Div B",
minDBMS:{ $min:"$Marks.DBMS"}}}]);
```

```
[ { _id: 'Div B', minDBMS: 40 } ]
```

#### **// Q3. Total SEPM marks of TE students in Pune**

```
test> db.student.aggregate([{$match:{Class:"TE", City:"Pune"}}, {$group:{_id:"Pune",
totalSEPM:{ $sum:"$Marks.SEPM"}}}]);
```

```
[ { _id: 'Pune', totalSEPM: 145 } ]
```

#### **// Q4. Division-wise count of TE students with DBMS marks 30–60**

```
test>db.student.aggregate([{$match:{Class:"TE", "Marks.DBMS":{ $gt:30, $lt:60 }}}, {
$group:{_id:"$Div", count:{ $sum:1}}}]);
```

```
[
  { _id: 'C', count: 1 },
  { _id: 'B', count: 2 },
  { _id: 'A', count: 1 }
]
```

#### **// Q5. Count of TE students from each city**

```
test>db.student.aggregate([{$match:{Class:"TE"}}, {$group:{_id:"$City", totalStudents:{
$sum:1}}}]);
```

```
[
  { _id: 'Mumbai', totalStudents: 2 },
  { _id: 'Pune', totalStudents: 2 }
]
```

#### **// Q6. Max SEPM marks among TE students city-wise**

```
test> db.student.aggregate([{$match:{Class:"TE"}}, {$group:{_id:"$City", maxSEPM:{
$max:"$Marks.SEPM"}}}]);
```

```
[ { _id: 'Mumbai', maxSEPM: 74 }, { _id: 'Pune', maxSEPM: 77 } ]
```

#### **// Q7. Compound Index (Name, Class)**

```
test> db.student.createIndex({ Name:1, Class:1 });
```

```
Name_1_Class_1
```

**// Q8. Single Index on Name**

```
test> db.student.createIndex({ Name:1 });
```

```
Name_1
```